Model monitoring pack: 12-month behavioural PD scorecard

Baseline: development sample (Dec 2015-2017). Tested: OOT Dec 2018 and monitoring Dec 2019 (COVID). RAG thresholds are the analyst's choice (industry rules of thumb).

Test	Thresholds (G / A / R)	OOT Dec 2018	RAG	Dec 2019	RAG
PSI, grade mix	<0.10 / .10-.25 / >.25	0.0018	Green	0.0077	Green
PSI, score deciles	<0.10 / .10-.25 / >.25	0.0012	Green	0.0064	Green
Gini (development 0.7973)	drop <10% / 10-20% / >20%	0.8430 (drop -5.7%)	Green	0.4759 (drop +40.3%)	Red
Grade back-test (worst grade)	p >=.05 / .01-.05 / <.01	G7 A1 R1	Red	G0 A0 R9	Red
Overall actual / expected	0.8-1.2 / 0.7-1.5 / outside	0.932 (0.449% vs 0.482%)	Green	7.535 (3.988% vs 0.529%)	Red
CSI, worst variable	<0.10 / .10-.25 / >.25	0.0667 (orig_rate)	Green	0.4881 (upb_ratio)	Red

Dec 2019 after adjusting for COVID forbearance (see page 7)

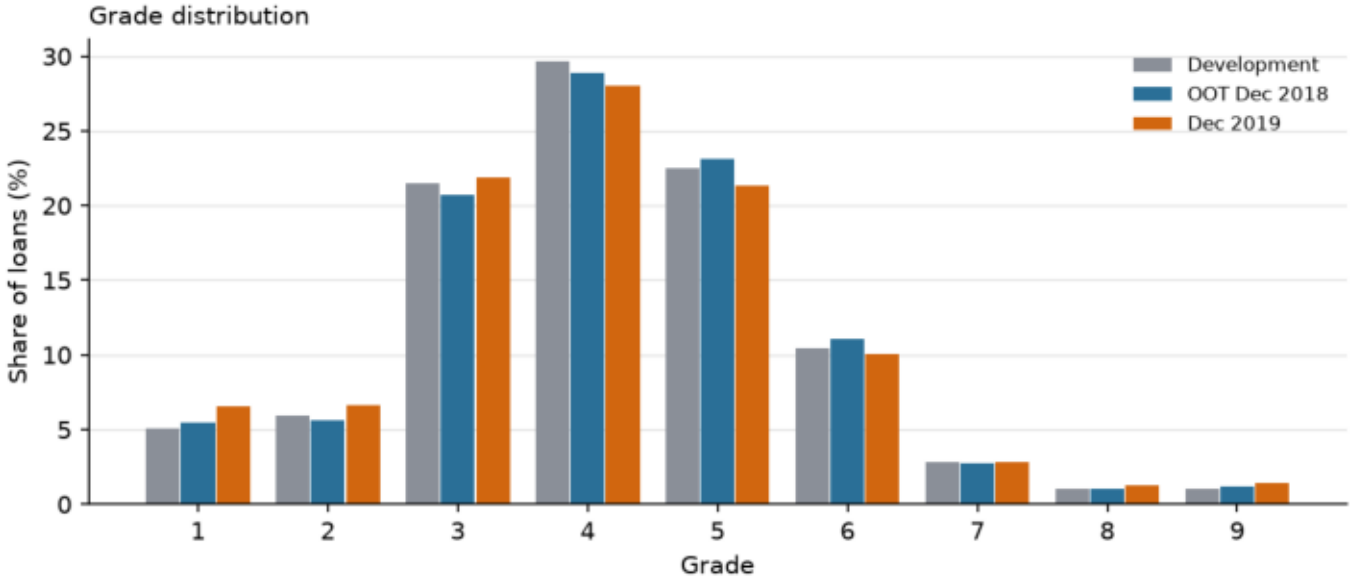
Test	Value	RAG
Dec 2019 Gini, forbearance-cured defaults treated as non-default	0.8049 (drop -0.9%)	Green
Dec 2019 actual / expected, same adjustment	0.857 (0.453% vs 0.529%)	Green
Dec 2019 grade back-test, same adjustment	G7 A1 R1	Red

Reading: the Dec 2019 failure is in the outcome, not the inputs. Population (PSI) is stable, but 93% of the 9,059 reported defaults were forborne loans that cured or paid off; only 136 ended in a credit-event termination. With forbearance-cured loans treated as non-defaults, Gini is back to 0.80 and actual/expected is 0.86. The grade back-test still shows grade 8 Red and grade 7 Amber, the same slope issue seen in OOT.

Test 1: Population stability (PSI)

GREEN

PSI = sum (A-E) ln(A/E); E = development share per grade / score decile, A = share now. Grade mix and score distribution are stable in both snapshots.

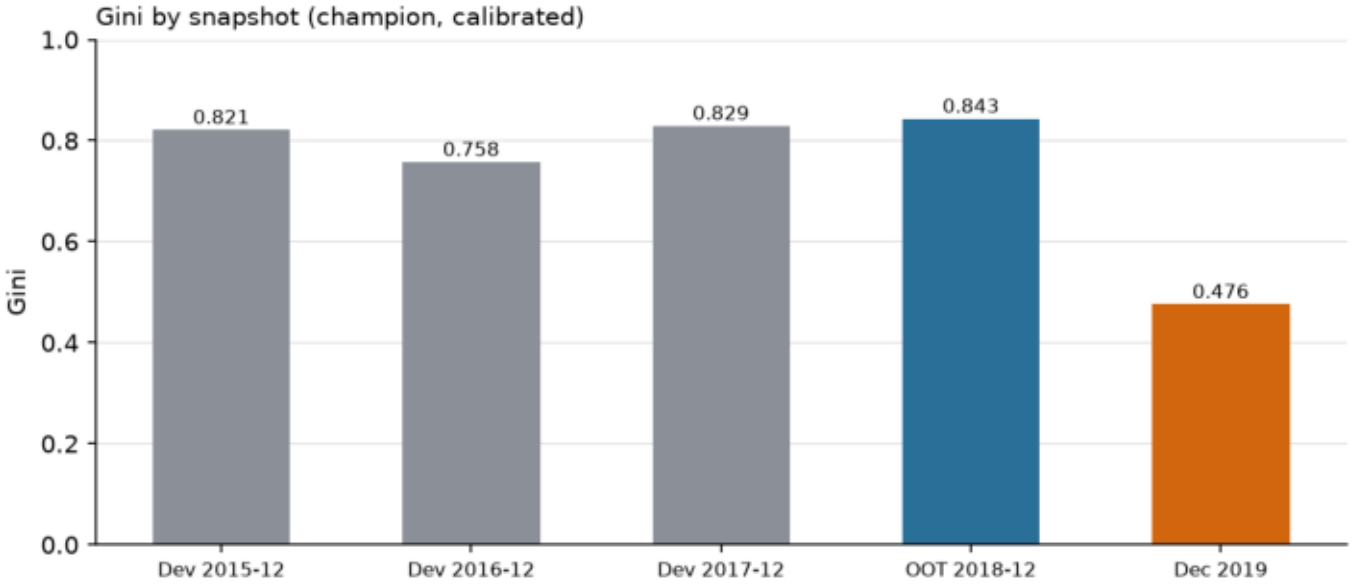


Sample	PSI grade	PSI score
OOT Dec 2018	0.0018	0.0012
Dec 2019	0.0077	0.0064

Test 2: Discrimination (Gini)

RED

Relative drop vs development Gini 0.7973. Thresholds: <10% Green, 10-20% Amber, >20% Red.

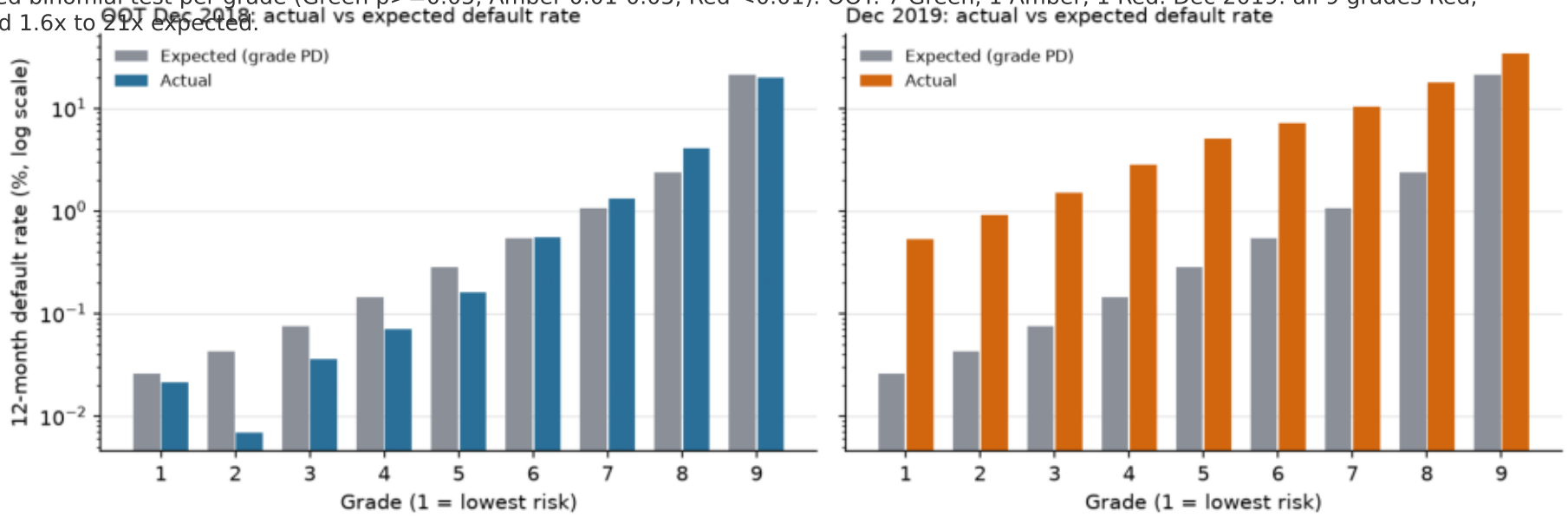


Snapshot	Gini	Loans
Dev 2015-12	0.8212	167,815
Dev 2016-12	0.7576	193,953
Dev 2017-12	0.8291	224,132
OOT 2018-12	0.8430	254,118
Dec 2019	0.4759	227,165
Dec 2019, forbearance-cured defaults as non-default	0.8049	227,165

Test 3: Grade back-test, actual vs expected by grade

RED

One-sided binomial test per grade (Green p>=0.05, Amber 0.01-0.05, Red <0.01). OOT: 7 Green, 1 Amber, 1 Red. Dec 2019: all 9 grades Red, observed 1.6x to 21x expected.



Grade	Grade PD	OOT observed	RAG	Dec 2019 observed	RAG
1	0.0261%	0.0215%	Green	0.5259%	Red
2	0.0428%	0.0070%	Green	0.9023%	Red
3	0.0745%	0.0361%	Green	1.4886%	Red
4	0.1453%	0.0708%	Green	2.8092%	Red
5	0.2804%	0.1597%	Green	4.9994%	Red
6	0.5395%	0.5525%	Green	7.1909%	Red
7	1.0639%	1.3201%	Amber	10.2665%	Red
8	2.3758%	4.0340%	Red	17.9359%	Red
9	21.3713%	19.7622%	Green	34.0452%	Red

Test 4: Overall actual vs expected default rate

RED

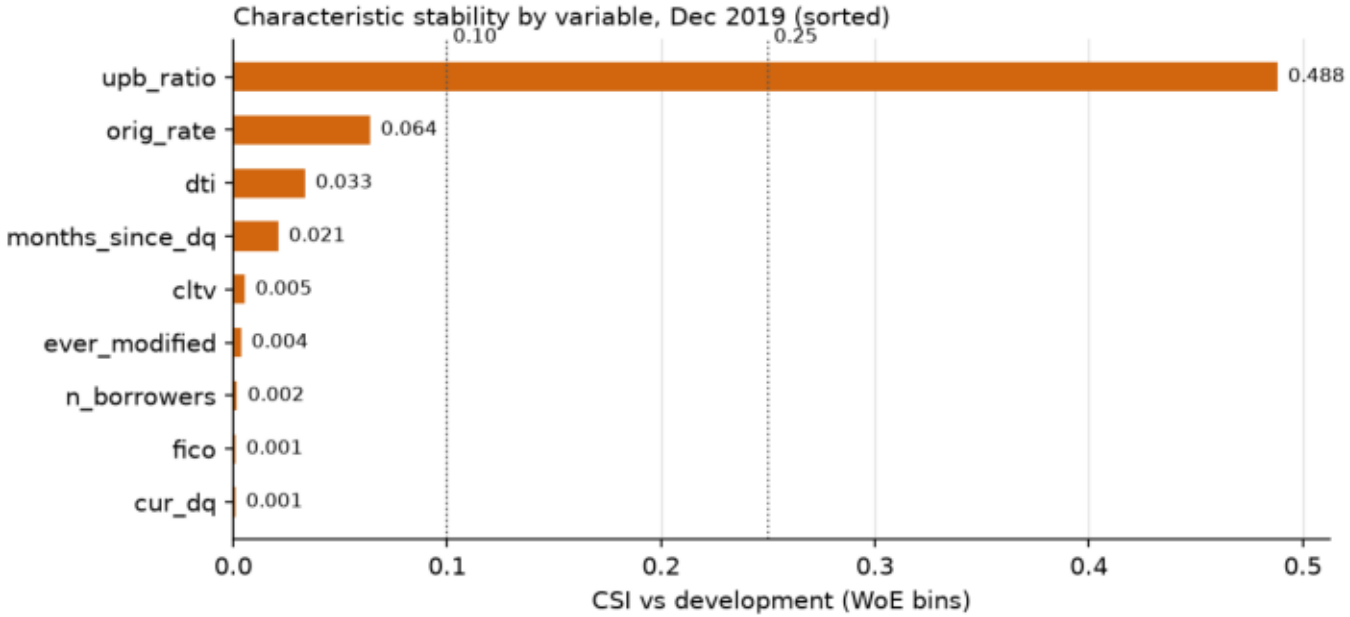
Thresholds: 0.8-1.2 Green; 0.7-0.8 or 1.2-1.5 Amber; outside Red.

Sample	Actual	Expected	A / E	RAG
OOT Dec 2018	0.4490%	0.4818%	0.932	Green
Dec 2019 (reported)	3.9879%	0.5292%	7.535	Red
Dec 2019 (forbearance-cured excluded)	0.4534%	0.5292%	0.857	Green

Test 5: Characteristic stability (CSI) per variable

RED

CSI = PSI on each variable's WoE bins vs development. Dec 2019 flags upb_ratio only (0.49): the sample has no 2019 originations, so the book seasons and pays down.



Variable	CSI OOT Dec 2018	CSI Dec 2019
upb_ratio	0.0576	0.4881
orig_rate	0.0667	0.0639
dti	0.0339	0.0335
months_since_dq	0.0043	0.0209
cltv	0.0058	0.0054
ever_modified	0.0021	0.0035
n_borrowers	0.0016	0.0017
fico	0.0014	0.0011
cur_dq	0.0002	0.0010

Finding: COVID forbearance and the Dec 2019 'defaults'

Loan-months Apr-Dec 2020 by assistance flag, and what happened to the 9,059 Dec 2019 defaults by end of data (Mar 2026).

Assistance flag	Loan-months	30+ DPD	90+ DPD
Forbearance plan (F)	72,608	86.1%	49.7%
No flag	1,665,823	0.9%	0.2%
Other assistance / deferral / disaster	15,407	32.6%	18.2%

Outcome of Dec 2019 default	Forborne	Not forborne	All
Credit-event termination (REO/short sale/charge-off/note sale)	61	75	136
Cured to current or paid off	8,029	448	8,477
Still delinquent / modified at end of data	376	70	446
All	8,466	593	9,059

Methodology and thresholds

- Baseline: development sample (train + holdout, Dec 2015-2017, 585,900 loan-snapshots).
- PSI on 9 grade shares and on development score deciles; CSI on each model variable's WoE bins.
- Gini drop is relative to the pooled development Gini.
- Back-test: one-sided binomial per grade against grade PD (development mean calibrated PD).
- RAG thresholds are common industry rules of thumb, chosen by the analyst, not regulatory values.
- Forbearance fields: Borrower Assistance Plan (F/T/R), Payment Deferral Flag (P/C), Delinquency Due to Disaster (Y).
- Outcomes after Dec 2020 use data to Mar 2026, i.e. hindsight a monitoring team would not have had at the time.